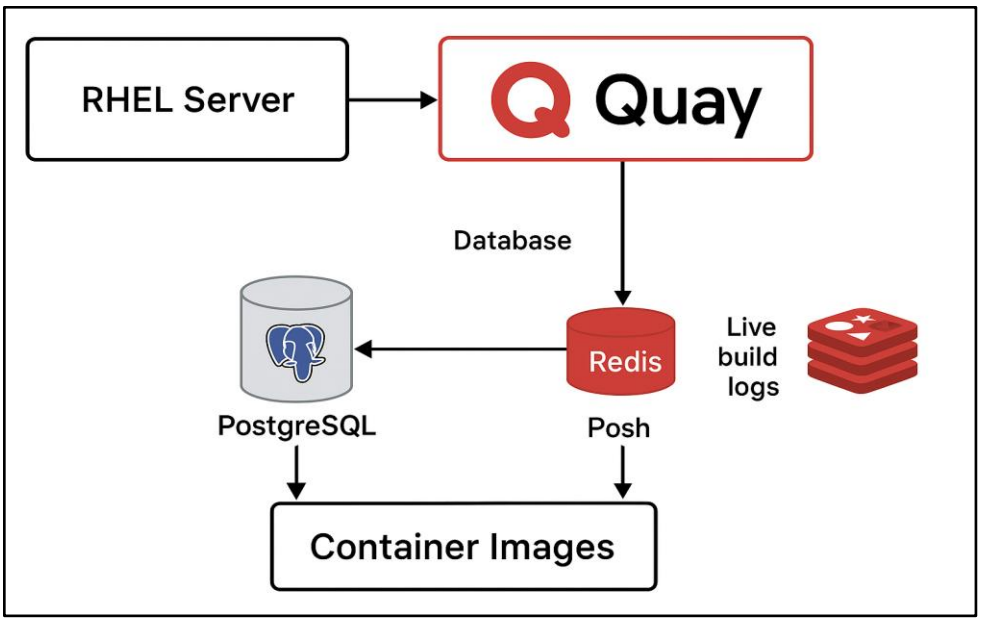
Setting Up Red Hat Quay on RHEL and Managing Container Images

Red Hat Quay is an enterprise-grade container registry used to store, secure, and manage container images. This guide walks through setting up Quay on a RHEL system, configuring required services, and adding container images.



1. Prerequisites

- **OS:** RHEL 9 (minimal or server+GUI)
- **Subscription:** Active Red Hat subscription
- **Hardware:**
    1. 2+ vCPUs
    2. 4 GB+ RAM
    3. ~30 GB Disk:

➔ 10 GB for RHEL OS
➔ 10 GB for containers (PostgreSQL, Redis, Quay)
➔ 10 GB for Quay local storage

- **Supported Architectures:** x86_64/amd64, s390x, ppc64le

2. Install and Register RHEL

- Update the system

```
sudo yum update -y
sudo yum install -y vim wget git net-tools firewalld
```

- Register the system

```
sudo subscription-manager register --username=<user> --
password=<password>
sudo subscription-manager refresh
Sudo subscription-manager repos --available
sudo subscription-manager attach --pool=<pool_id>
```

- Enable repositories

```
sudo subscription-manager repos --enable "rhel-9-for-x86_64-
appstream-rpms"
sudo subscription-manager repos --enable "rhel-9-for-x86_64-baseos-
rpms"
```

- Install Podman
  Podman is used to run containers without a separate daemon.

```
sudo yum install -y podman
```

or full container tools

```
sudo yum module install -y container-tools
```

- Configure Firewall
  Open required ports for Quay, PostgreSQL, and Redis:

```
sudo firewall-cmd --permanent --add-port=80/tcp
sudo firewall-cmd --permanent --add-port=443/tcp
sudo firewall-cmd --permanent --add-port=5432/tcp
sudo firewall-cmd --permanent --add-port=6379/tcp
sudo firewall-cmd --reload
```

- Networking & Hostname

Set hostname and configure /etc/hosts:

```
sudo hostnamectl set-hostname quay-server.example.com
Vim /etc/hosts
```

- Replace 192.168.95.111 with your server's IP.
- Containers will use this hostname to communicate with the registry, database, and Redis.

```
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.95.111 quay-server.example.com
```

Start PostgreSQL Container

- Quay requires a PostgreSQL database.

1. Create storage directory:

```
mkdir -p /quay/postgres-quay
setfacl -m u:26:-wx /quay/postgres-quay
```

2. Run PostgreSQL container:

```
sudo podman run -d --name postgresql-quay \
  -p 5432:5432 \
  -e POSTGRES_USER=quayuser \
  -e POSTGRES_PASSWORD=quaypass \
  -e POSTGRES_DB=quay \
  -e POSTGRES_ADMIN_PASSWORD=adminpass \
  -v /quay/postgres-quay:/var/lib/pgsql/data:Z \
  registry.redhat.io/rhel8/postgresql-13:latest
```

3. Install pg_trgm extension (required by Quay):

```
sudo podman exec -it postgresql-quay \
  /bin/bash -c 'echo "CREATE EXTENSION IF NOT EXISTS pg_trgm" | psql -
d quay -U postgres'
```

Start Redis Container

Redis is required for live build logs and user events.

```
sudo podman run -d --name redis \
  -p 6379:6379 \
  -e REDIS_PASSWORD=redhat \
  registry.redhat.io/rhel8/redis-6:1-110
```

- Use a strong password for production.
- Note the host and port (quay-server.example.com:6379) for configuration.

Configure Quay

- Create Quay Config Directory

```
mkdir -p /quay/config
```

- Create YAML Configuration

```
[root@quay-server ~]# vim $QUAY/config/config.yaml
```

```yaml
BUILDLOGS_REDIS:
    host: quay-server.example.com
    password: redhat
    port: 6379

CREATE_NAMESPACE_ON_PUSH: true
DATABASE_SECRET_KEY: a8c2744b-7004-4af2-bcee-e417e7bdd235
DB_URI: postgresql://quayuser:quaypass@quay-server.example.com:5432/quay

DISTRIBUTED_STORAGE_CONFIG:
    default:
        - LocalStorage
        - storage_path: /datastorage/registry
DISTRIBUTED_STORAGE_DEFAULT_LOCATIONS: []
DISTRIBUTED_STORAGE_PREFERENCE:
    - default

FEATURE_MAILING: false
SECRET_KEY: e9bd34f4-900c-436a-979e-7530e5d74ac8

SERVER_HOSTNAME: quay-server.example.com
SETUP_COMPLETE: true

USER_EVENTS_REDIS:
    host: quay-server.example.com
    password: redhat
    port: 6379

SUPER_USERS:
  - quayadmin
```

- Replace redhat with your Redis password.

- SUPER_USERS are admins who can manage users, organizations, logs, and keys.

Set Up Quay Superuser Authentication

1. Create htpasswd file:

```
sudo htpasswd -B -c /quay/config/htpasswd quayadmin redhat
```

- -B uses bcrypt for secure password storage.
- This sets password redhat for user quayadmin.

Create Storage for Quay Images

```
mkdir -p /quay/storage
setfacl -m u:1001:-wx /quay/storage
```

Run Quay Container

```
sudo podman run -d --name quay \
  -p 80:8080 -p 443:8443 \
  -v /quay/config:/conf/stack:Z \
  -v /quay/storage:/datastorage:Z \
  registry.redhat.io/quay/quay-rhel8:v3.15.1
```

- Mount /quay/config for configuration.
- Mount /quay/storage for container images.
- Ports 80 and 443 provide access to Quay Web UI.

Verify Quay Startup

```
sudo podman logs quay
```

- Look for errors in the database or Redis connectivity.
- Ensure Validating Configuration reports all checks as successful.

Login to Quay

```
sudo podman login --tls-verify=false quay-server.example.com
```

- Enter quayadmin and the password set in htpasswd.
- Login should succeed.

Add Container Images

- BusyBox Example

1. Pull local image:

```
sudo podman pull docker.io/library/busybox:latest
```

- Enter the following command to see the local copy of the image

```
sudo podman images
```

```
REPOSITORY                                      TAG         IMAGE ID      CREATED        SIZE
quay-server.example.com/quayadmin/ubi           latest      4461e0f8948d  13 hours ago   217 MB
registry.access.redhat.com/ubi9/ubi             latest      4461e0f8948d  13 hours ago   217 MB
registry.redhat.io/rhel8/postgresql-13          latest      c28fc91f8e0c  5 days ago     506 MB
quay-server.example.com/quayadmin/ubuntu        latest      97bed23a3497  12 days ago    80.6 MB
docker.io/library/ubuntu                        latest      97bed23a3497  12 days ago    80.6 MB
registry.redhat.io/quay/quay-rhel8              v3.15.1     038250f3aace  2 months ago   1.2 GB
quay-server.example.com/quayadmin/busybox       test        0ed463b26dae  12 months ago  4.68 MB
docker.io/library/busybox                       latest      0ed463b26dae  12 months ago  4.68 MB
registry.redhat.io/rhel8/redis-6                1-110       51732cea2f83  2 years ago    300 MB
```

2. Tag for Quay registry:

```
sudo podman tag busybox:latest quay-
server.example.com/quayadmin/busybox:test
```

3. Push to Quay:

```
sudo podman push --tls-verify=false quay-
server.example.com/quayadmin/busybox:test
```

4. Pull back to test:

```
sudo podman rmi quay-server.example.com/quayadmin/busybox:test
sudo podman pull --tls-verify=false quay-
server.example.com/quayadmin/busybox:test
```

- Ubuntu Example

1. Pull Ubuntu image:

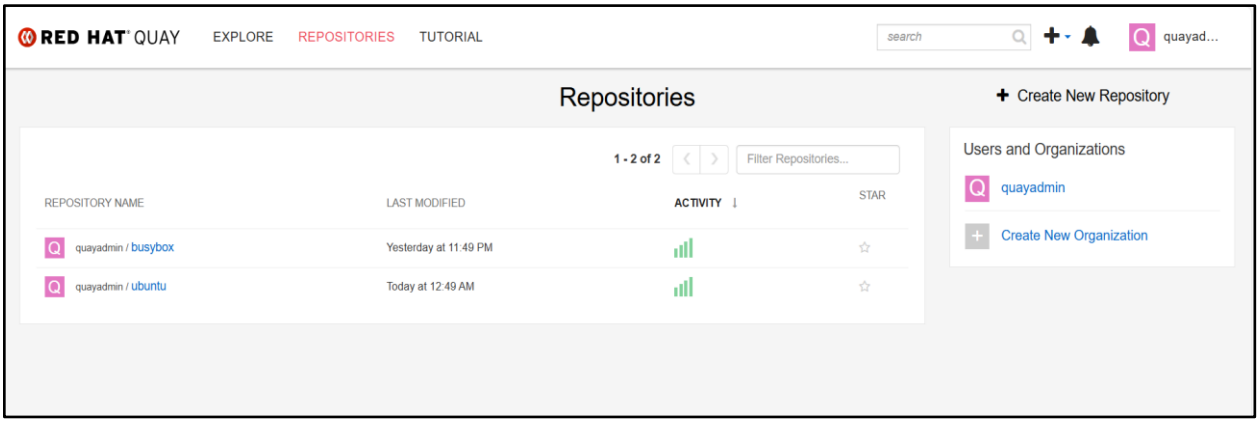
```
sudo podman pull docker.io/library/ubuntu:latest
```

2. Tag for Quay:

```
sudo podman tag ubuntu:latest quay-
server.example.com/quayadmin/ubuntu:latest
```

3. Push to Quay:

```
sudo podman push --tls-verify=false quay-
server.example.com/quayadmin/ubuntu:latest
```

Access Quay Web UI

- Open browser:http://192.168.95.111/repository/
- Log in with quayadmin credentials.
- Superusers can manage users, repositories, service keys, and logs.